

Root sprouting in American beech: production, survival, and the effect of parent tree vigor

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Root sprout age-class distributions around American beech trees were measured to characterize production of sprouts under closed canopies. Annual mortality of root sprouts was estimated by static and cohort life table analyses. Sprouts around parent trees with and without beech bark disease were compared to test for effects of lowered parent vigor on sprout production and vigor. Age-class distributions were highly variable, indicating episodic production of sprouts. Trends in the data suggested that (i) for individual parent beech trees, the number of sprouts per age-class decreased exponentially as sprout age increased; and (ii) parent trees with larger diameters had more sprouts, more sprout age-classes, but greater variability in age-class distribution. Life table analyses indicated uniform per capita mortality rates for clumps of sprouts but decreasing mortality with age for individual sprouts within clumps. Low parent vigor, due in part to beech bark disease, was weakly correlated with reduced sprout production, but diseased trees maintained populations of older sprouts that differed little from sprouts associated with nondiseased trees.

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Les auteurs ont étudié la distribution d'âges des rejets racinaires produits par le hêtre à grandes feuilles afin d'en caractériser la production sous couvert fermé. Le taux de mortalité annuel des rejets a été estimé à l'aide de tables de survie. On a comparé les rejets autour d'arbres parents touchés ou non par la maladie corticale du hêtre pour évaluer l'effet de la vigueur du parent sur la production et la vigueur des rejets. La distribution très variable des classes d'âge suggère une production irrégulière de rejets. Les données tendent à montrer que: (i) pour chacun des arbres, le nombre de rejets par distribution d'âges diminue de façon exponentielle avec l'âge des rejets; et (ii) les arbres avec les plus gros diamètres produisent des rejets comportant une plus grande distribution d'âges avec une plus grande variabilité dans la distribution des classes d'âge. L'analyse des tables de survie indique que le taux de mortalité des rejets est uniforme d'un arbre à l'autre mais qu'il diminue avec l'âge des rejets dans un même groupe. Le manque de vigueur des arbres atteints de la maladie corticale est faiblement corrélé à une production moins abondante de rejets. Par contre, les arbres malades ont autant de rejets plus âgés que les arbres sains.

[Traduit par la revue]

Introduction

Successful regeneration in the shade-tolerant species *Fagus grandifolia* Ehrh. (American beech) depends on establishment and survival of seedlings and root sprouts under closed canopies (Fowells 1965). Studies have shown that beech seed production and seedling establishment are heavy in some years but light in others (Fowells 1965; Dix and Skrentny 1965; Forcier 1973) and that per capita mortality of beech seedlings in the understory decreases with time (Forcier 1973; Lenk 1982). No published studies, however, detail the temporal patterns of production and survival among root sprouts in American beech. Since sprouts may retain functional connection to parents for many years, as evidenced by movement of herbicides from parent stems to adjacent sprouts (Rushmore 1956; Abrahamson 1983), it is likely that production and survival of beech sprouts and seedlings are different.

The development of beech bark disease, a decline disease affecting beech in much of its natural range, affords both the opportunity and need to characterize effects of parent vigor on sprouting. The disease is initiated by a scale insect, *Cryptococcus fagisuga* Lind., which is followed by cambium killing fungi, *Nectria* spp. (Ehrlich 1934). Although infection occurs in all size classes of beech, most disease-caused mortality occurs in trees greater than 25 cm diameter at breast height (dbh) (Mize and Lea 1979; D. R. Houston, personal communication). Shigo (1972) identified three stages in regional disease development: (i) an advanced front where recently established popula-

tions of scale are frequent; (ii) a killing front where infection by fungi is causing substantial mortality of large trees; and (iii) an aftermath stage where disease incidence, characterized by cankering defect, is widespread, mortality is low, and the scale is now endemic. Observations of dense beech reproduction in stands having heavy mortality have led to speculation that sprouting is stimulated by parent decline or by harvesting trees before they die (Houston 1975). However, in one study of stands in the aftermath stage, little change in the abundance of beech regeneration (seedlings and sprouts combined) was noted in stands where the disease caused slow decline and mortality of scattered individuals (Twery and Patterson 1984).

In the present study, we characterize temporal patterns of sprout production and survival by determining age-class distributions of sprouts around individual parent beech trees found in closed-canopied forests, and by measuring vigor and survival of a subset of these sprouts over 2 years. Sprout age-class distributions around diseased and nondiseased parent trees are then compared to infer effects of lowered parent tree vigor on sprout production and survival.

Methods

Sprout production

Thirty-one parent trees, some heavily diseased and some without beech bark disease, were located for sampling in central New York State. The trees were in 13 closed-canopied northern hardwood and northern hardwood-conifer stands (one to seven trees per stand) that were unlogged in the last 20 or more years. Predominant species in the stands were sugar maple (*Acer saccharum* Marsh.), red maple, (*Acer rubrum* L.), black cherry (*Prunus serotina* Ehrh.), eastern hemlock

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(*Tsuga canadensis* (L.) Carr.), American beech, and white ash (*Fraxinus americana* L.). The killing front of the disease passed through the area in the 1970's (Miller-Weeks 1983) causing only scattered mortality in the sampled stands.

Within square sample plots² (24 × 24 m) centered on each parent, all beech stems were mapped and assigned to one of three origins (seedling, sprout, or unknown) based on root collar excavation. Because parent trees were at least 20 m away from any other beech greater than 10 cm dbh, parental affiliation of sprouts was readily determined. For all sprouts, total height was measured and age was determined by counting bud scale scars or by sectioning the base for xylem ring counts. Sprouts were often clumped at a single point on a parent root. In these cases, height and age were determined for the tallest sprout in the clump only.

To determine if parent tree size or if environmental factors most strongly affect sprout production, a stepwise multiple linear regression technique (SAS Institute Inc. 1982) was used to test for relations between the total number of sprouts in the plot as the dependent variable and (i) parent size measures including dbh, total height, basal area (dbh²), and approximate biomass from a regression developed by Monteith (Tritton and Hornbeck 1982) that incorporates dbh, total height, and dbh²; and (ii) environmental factors including plot slope (%), plot aspect (degrees from due north), total plot basal area (woody stems 3 cm or larger in dbh), basal area of sugar maple in the plot, and basal area of eastern hemlock in the plot.

Temporal patterns of sprout production (e.g., whether sprouts are produced every year or sporadically over the years) was inferred from the number of 1-year-old sprouts and from the presence of large numbers of sprouts in older age-classes.

Sprout survival

Sprout survival was investigated by use of both static and cohort life table analyses (Hett and Loucks 1976; Krebs 1985). Static life table analysis incorporated the combined sprout age-class distribution of all 31 parent trees (1057 sprouts) to infer sprout survival, assuming constant annual production and age-specific mortality. For the cohort life table, 331 sprouts around 21 of the 31 trees were inventoried in mid-summer 1983 and 1984 and re inventoried during late summer in 1984 and 1985. During re inventory, the tallest sprout in each clump was assigned to one of the following vigor categories: (i) vigorous (more than 30 cm of current annual shoot growth); (ii) healthy; (iii) some dieback (some or all of current year's growth dead); (iv) significant dieback (some of past year's growth dead); and (v) dead. The number and survival of other sprouts within each clump was also recorded.

Effects of parent tree vigor

For each parent, two direct measures of parent vigor were made: radial growth and crown health index. Radial growth for each of the last 20 years was estimated from five increment cores taken at 1.37 m height. Crown health was assigned an index value of 0 (healthy), 1 (chlorotic or with a few minor branches recently dead), 2 (major branches recently dead but more than 50% of the crown still alive), or 3 (more than 50% of crown recently dead). Other measures made for each parent included dbh, total height, percent cover of the bole by *Nectria* spp. perithecia, and percent cover of the bole by the wax of *Cryptococcus fagisuga*.

No natural groupings of parent trees were noted when radial growth was plotted against crown health (except one group of two trees growing much faster in radius than any of the others). Consequently, parent trees were divided arbitrarily into three vigor classes as follows: (i) the 31 trees were ranked twice, once according to radial growth in the 3 years prior to coring and once according to crown health index (highest vigor achieved lowest rank and tied values were given mean rank); (ii) the two ranks were summed for each tree; and (iii) the 31 sums were divided into the lowest 10, intermediate 11, and highest 10

to form high-, intermediate-, and low-vigor parent classes, respectively. Based on results of the above classification, values of crown health index and radial growth that separate the classes were calculated and are shown in Table 1. Relationships among and within the two direct measures of parent vigor and between these and other tree attributes were investigated using Spearman's rank correlation statistic (see e.g., Sokal and Rohlf 1981).

To investigate the effects of low parent vigor on sprout production, we first compared sprout age-class distributions to see if certain distribution types, such as inverse j-shaped, bell-shaped, or uniform, were consistently associated with particular parent vigor classes. We then combined sprout age-class distributions within the three parent vigor classes and used a G-test (see e.g., Sokal and Rohlf 1981) to see whether the three distributions were identical. Since sprout age-class distributions are the result of sprout production and survival over many years, while parent vigor was a measure of current but not necessarily past parent physiological status, we focused our comparison on sprouts in young age-classes.

To determine if parent vigor is related to sprout vigor or survival, re inventoried sprouts were grouped into those around high- and low-vigor parent trees, and similarity between the two groups in the proportions of sprouts in each of the five sprout vigor categories was assessed using the G-statistic.

Results and discussion

Sprout production

Total number of sprouts per clone was related to all parent size measures and most strongly to parent basal area. After the effect of basal area was taken into account, dbh, total height, and approximate biomass were no longer significant. There was no evidence that any of the environmental factors had an effect. Consequently, the regression function of total number of sprouts per clone (Y) and parent tree basal area in m² (BA) took the following form:

$$[1] \quad Y = 4.4894 + 268.1195 \text{ BA}$$

with an R^2 of 0.36 ($df = 1/29$, $p = 0.0003$). Thus, larger trees produce more sprouts on average (assuming that sprout survival is not affected by tree size). That plot slope, aspect, and tree composition were poor predictors of sprouting is consistent with data from other studies showing little or no correlation of environment factors with the density of beech sprouts (Lenk 1982; Held 1983).

Age-class distributions of sprouts arranged in order of parent dbh are shown in Fig. 1. The data show considerable variation in sprout age-class distributions among trees, even among trees within the same stand. This was due in part to large variation in total number of sprouts per parent (mean = 34.1, SD = 27.12, minimum = 3, maximum = 110) and in part to episodes of sprout production followed by some years of much reduced production. Episodes of heavy or light production were apparently not synchronized among clones as seen in the lack of peak or valley alignment for individual sprout age-classes across clones (Fig. 1). The nonlinear regression equation

$$[2] \quad N = 0.60223e^{-0.1218(A) + 0.0521(D)}$$

where N is number of sprouts, A is sprout age in years and D is parent dbh, fits through the center of the data in Fig. 1 (sum of residuals was -0.334 sprouts). This regression reflects a trend for exponential declines in sprout-age class distributions for each individual tree (Fig. 2). Further, it reveals that more sprouts and more sprout age-classes occur as parent dbh increases.

The distribution of sprout heights was similar to that for

²Square plots were used to facilitate grid layout and mapping. The plot dimensions were larger than needed to include all sprouts associated with the parent tree (Jones and Raynal 1986).

TABLE 1. Values of crown health index and 3-year radial growth separating sampled parent trees into three vigor classes

Vigor class	Crown health index	3-year radial growth (mm)
High	0	>1.00
	1	>2.40
Intermediate	0	0.55–1.00
	1	1.50–2.40
	2	>2.35
Low	0	<0.55
	1	<1.50
	2	<2.35
	3	all

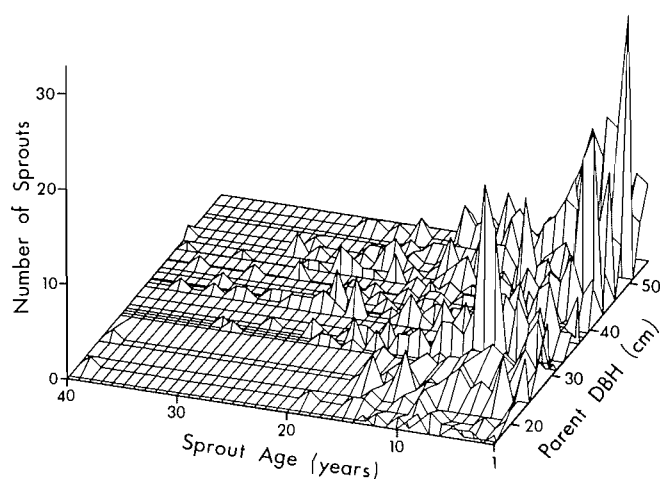


FIG. 1. Observed sprout age-class distributions by parent dbh.

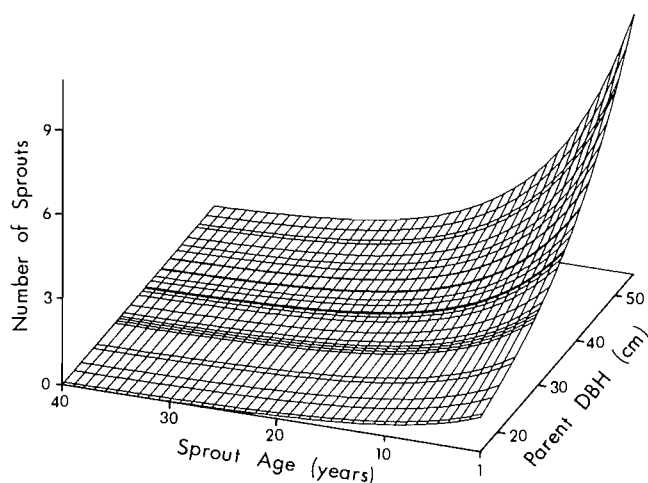


FIG. 2. Predicted sprout age-class distribution by parent dbh using equation 2.

sprout age (Fig. 3). There were many sprouts of short stature and decreasing numbers of sprouts in taller height classes.

Sprout survival

Using static life table data, we found a trend for linear decline when $\log_{10}$ (number of sprouts) was plotted over sprout age (Fig. 4). Such a decline indicates uniform per capita mortality.

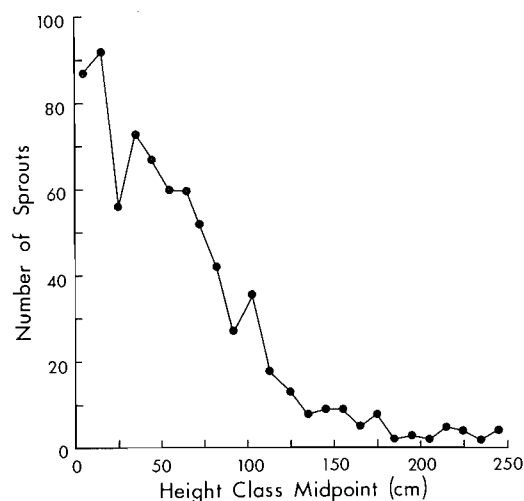


FIG. 3. Height class distribution of sprouts from all 31 clones combined.

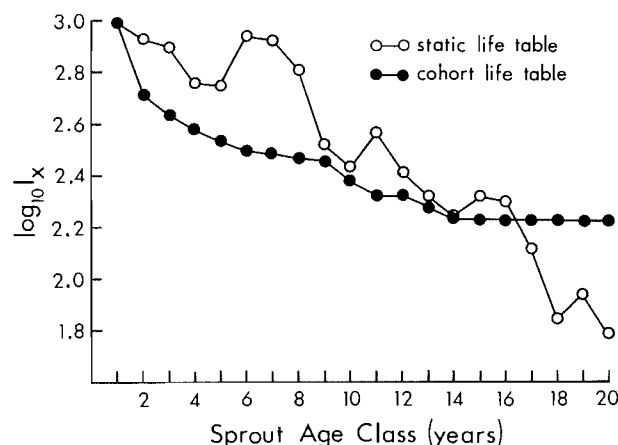


FIG. 4. Sprout survivorship curves estimated by static and cohort life table analyses.

Prior to cohort life table analysis, sprouts around two trees were excluded because the parent stems snapped off in winter 1983–1984. One of the trees was classified as low in vigor and had unusually high sprout mortality (63.0% annually versus 9.3% for trees still having living crowns). The other was classified as intermediate in vigor and had 11.1% annual mortality of reinventoried sprouts.

The plot of cohort life table data indicated a steady decline in per capita mortality rate from age 1 to age 14, with no mortality noticed after age 14 (Fig. 4). For all reinventoried stems within clumps (including those not the tallest), annual mortality was 30.0%, and much of this was concentrated in 1- and 2-year-old stems according to field notes (high mortality of these age classes was also noted among the tallest stems; Fig. 4). Compared with individual stems, clumps had lower mortality rates. The number of clumps having all stems die and no new ones produced was only 5.7%. Thus, turnover of stems within clumps was faster than turnover of clumps within the clone. Since high mortality of young individual sprout stems is buffered by the tendency for higher numbers of stems in young sprout clumps (see Jones and Raynal 1986), clumps have more constant per capita mortality than do individual stems. Therefore, we assume that the static life table data provides a reasonable approximation of true sprout clump mortality, and we use this data below in further analyses.

TABLE 2. Rank correlation of vigor measures among 31 trees

Vigor measure or tree attribute	Vigor measure			
	Crown health	3-year radial growth	% fungus	% scale
Crown health	—			
3-year radial growth	0.11 (0.559)	—		
% fungus	0.58 (0.001)	0.09 (0.619)	—	
% scale	0.30 (0.098)	0.26 (0.155)	0.36 (0.046)	—
dbh	0.27 (0.139)	0.28 (0.121)	0.20 (0.275)	0.19 (0.319)
Total height	-0.10 (0.599)	-0.19 (0.308)	0.08 (0.671)	0.15 (0.436)

NOTE: Probability of greater correlation coefficients under the hypothesis of no correlation are shown in parentheses.

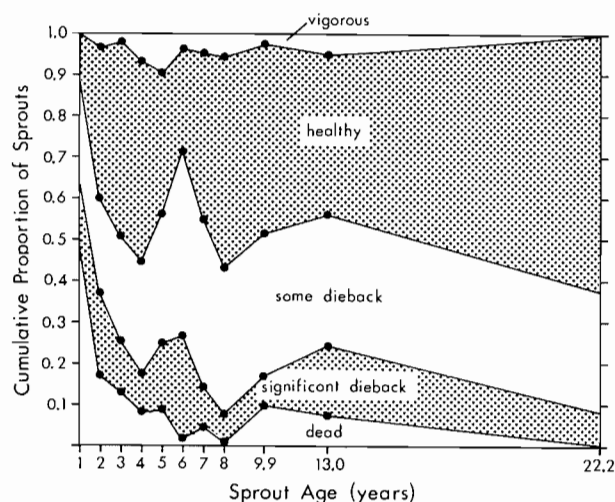


FIG. 5. Proportions of sprouts in five vigor categories based on 331 sprouts reinventoried in 1984 and 1985.

The following nonlinear exponential regression equation was fitted to the static life table curve to provide an estimate of sprout half-life which can be used in turn to estimate mean sprout life expectancy and annual per capita mortality:

$$[3] \quad N_i = B(1/2)^{A/T} + e_i$$

where N_i is the number of sprouts in age class i , B is a location parameter, A is sprout age in years, T is half-life of sprouts (from Crowe and Crowe 1969), $T/\ln 2$ is mean sprout life expectancy (from Meyer 1970), $\ln 2/T$ is annual per capita mortality, and e_i is error. The resulting estimate of sprout clump half-life was 6.2 years with an approximate standard error of 0.58 year. Estimated mean life expectancy was 8.9 years and annual per capita mortality was 11.3%. Such life expectancies may account for the continuous presence of sprouts in the sampled clones despite episodic sprout production.

The proportions of reinventoried sprouts in five vigor classes are shown in Fig. 5. Due to a low number of sprouts greater than 9 years old, data for ages 8–11 (mean age = 9.90 years) were pooled into one age-class as were ages 12–15 (mean age = 13.02 years) and more than 15 (mean age = 26.22 years). Between ages 1 and 4, the proportion of dead and dying sprouts declined. Beyond age 4, proportions in all vigor classes remained relatively steady except for an unusually low proportion of healthy sprouts at age 6. The data show that at least half of the sprouts had some sort of dieback each year, a finding not unexpected for trees growing under closed canopies.

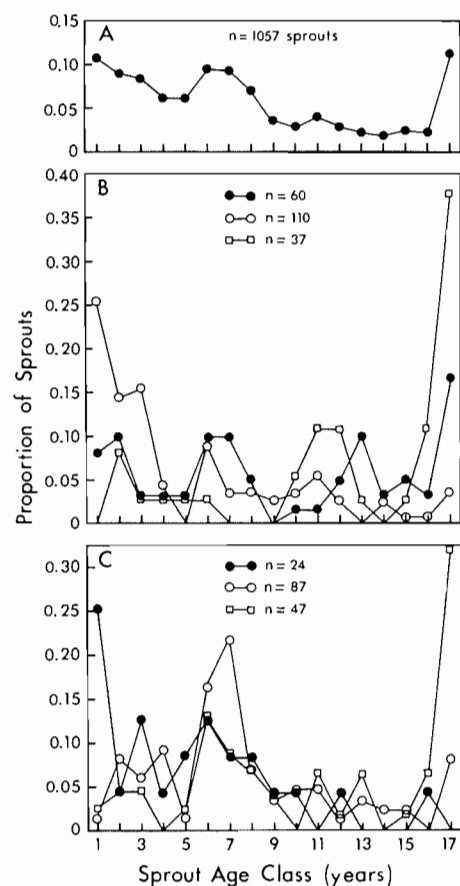


FIG. 6. Proportion of sprouts in 17 age-classes: (A) 31 trees combined, (B) 3 representative high-vigor parent trees, (C) 3 representative low-vigor trees. n = number of sprouts.

Effect of parent vigor

A correlation between crown health index and percent *Nectria* cover ($r = 0.58$, $df = 29$, $p = 0.001$) and a weaker correlation between percent *Nectria* and percent *Cryptococcus* ($r = 0.36$, $df = 29$, $p = 0.046$) were the only significant rank correlations noted among parent tree vigor measures (Table 2). The lack of strong correlations is indicative of a decline disease where many factors interact to affect tree vigor. That tree vigor measures were more strongly correlated with abundance of *Nectria* than with abundance of *Cryptococcus* was expected since the fungus is more injurious to the tree.

Within the three parent vigor classes, no distinct pattern of age-class distribution could be identified as shown by curves for

TABLE 3. Vigor categories for reinventoried sprouts in two groups: sprouts around high- and low-vigor trees

Sprout vigor category	High-vigor clones (10)		Low-vigor clones (5)	
	No. of observations	%	No. of observations	%
Vigorous	11	4.40	6	4.41
Healthy	97	38.80	78	57.35
Some dieback	87	34.80	32	23.53
Significant dieback	34	13.60	13	9.56
Dead	21	8.40	7	5.15
Total	250	100.00	136	100.00

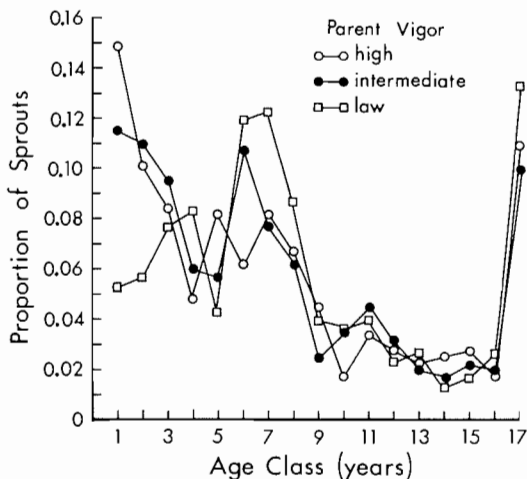


FIG. 7. Sprout age-class distribution for high, intermediate, and low parent vigor classes.

representative trees in Fig. 6. When sprout age-class distributions within each parent vigor class were combined (Fig. 7) and then compared, the proportional distributions differed ($G = 54.2$, $df = 32$, $p = 0.008$). When only the high- and low-vigor parent classes were so compared, there was weaker evidence that patterns of sprout age-class distributions differed ($G = 26.2$, $df = 16$, $p = 0.051$). Much of the difference between distributions was attributable to the 1-, 2-, and 7-year-old age-classes (Fig. 7); however, tests for differences in these specific age-classes cannot be made after all age-classes are used in the G-test. Low vigor trees had relatively low proportions of 1- and 2-year-old but high proportions of 7-year-old sprouts. Thus, our data show an apparent drop in production of new sprout clumps when parent trees lose vigor. A trend for decreased production of individual stems within young sprout clumps was also noted around low-vigor parent trees (Jones and Raynal 1986). The high proportion of 7-year-old sprouts in the low-vigor trees is not easily explained; however, most trees classified as low vigor had much higher radial growth rates 7 years ago implying a higher vigor status in the past.

Low-vigor trees had somewhat healthier sprouts ($G = 12.84$, $df = 4$, $p = 0.012$), but differences in observed sprout mortality were minor (Table 3). Greater health among sprouts around low-vigor clones may be due to many factors, including increased light levels in the understory as the parent tree crown begins to deteriorate, indirect effects of changed light environments such as altered levels of herbivory or spring frost damage, and changes in carbon allocation patterns due to interactions with the disease.

Although lowered parent vigor apparently had minor or no effect on sprout production and survival, our data only cover clones with living parents. The high mortality of sprouts noted in the low-vigor clone that snapped off may indicate an overall adverse effect of beech bark disease on long-term sprout survival. In general, our findings agree with those of Twery and Patterson (1985) who found little or no decline in beech regeneration in stands having slow and scattered development of beech bark disease.

Conclusions

American beech trees produce populations of root sprouts that survive in the understory for many years. For individual parent trees, annual sprout production is highly variable and not synchronized with production in nearby parents. On average, larger parent trees produce more sprouts. Sprout survival is apparently constant across sprout age-classes and is high enough to maintain sprout populations during years when none are produced. Reduced parent vigor has little effect on sprout production and survival as long as the parent tree remains alive.

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